RD-03 C0 Gate v2: Mesh-Convergence Study

FEMM/MATLAB extraction-and-solver verification rebuilt to the SOL-RD03-C0e-R stop-ship list: manufactured-solution operator certification, five-mesh coax known-answer family with co-refined arc discretization, four-mesh axisymmetric loop family against elliptic-integral quadrature references, Richardson/GCI machinery meta-verified against exact solutions, and the FEMM convention set locked by experiment. Family-agnostic: relationships and methods only, no family numbers, no part numbers, no prices.

Field	Value
Milestone	C0 verification gate v2 — ALL GATES PASS. Coax energy in the asymptotic range (ratio 1.0004); observed orders stable; GCI machinery validated against exact solutions; retraction R-b settled by a nonzero complex known-answer test.
Built / verified	2026-07-22 19:09–20:12 America/Chicago - MATLAB R2026a U3 (terminal-launched batch, durable logs) - FEMM 4.2 21Apr2019 x64 (femm.exe SHA-256 matches the C0e-R audit record).
Run ID / cache key	run_20260722_200912 · cache key 358b5b996d40eedd… (hash of all sources + femm.exe + MATLAB version + parameters)
Repo	datum-transmissions: verification/c0_v2/ (9 sources) · reports/c0_v2/ (figures, run JSON, transcript, gate manifest)
Verification	Stage A′ 3/3 functionals order 2.00 · coax 4/4 gate criteria · loop 2/2 · transcript 2,747 B (non-empty, asserted)
Authority	Verification-layer evidence ONLY. No application adequacy is claimed: decision-layer thresholds await the S2b error budget [NISTIR8298 posture]. Every value below is [Measured] from this run unless tagged otherwise.

01 / Scope and posture

What this buys Datum: a trustworthy FEMM+MATLAB verification gate is the prerequisite for the C0 dynamic-magnetics decision (eddy branch vs. diffusion state) that RD-03P's dynamic-current screen depends on. Two prior FEMM campaigns (v2–v5) were lost to extraction defects that this gate class would have caught; the cost of NOT having it is already measured.

Sol 5.6's adversarial audit (SOL-RD03-C0e-R, 2026-07-22) returned HOLD with five stop-ship defects and a corrected design for the mesh study. This report is the rebuilt gate, rerun from scratch, with every P0 repair landed and the audit's Q1–Q8 design answers folded in. It is written for Sol's re-audit: every claim carries its number, every number carries a sample calculation in Appendix A, and the full evidence bundle (sources, .fem/.ans, transcript, manifest with hashes) is retained in an immutable run directory.

Two-layer threshold policy adopted from the audit (Q3): the **verification gate** (this report) uses exact/manufactured targets, observed-order behavior, and asymptotic-range checks; the **decision gate** (pass/fail against donor requirements) is NOT exercised — it requires the S2b error budget, which is blocked on shop data only Mohammed can supply.

02 / Corrections adopted from SOL-RD03-C0e-R

Audit item	Disposition in v2
P0-1 fail-closed runner	Every stage gate computed, persisted, then ASSERTED. gate_manifest.json written with pass bits + cache key; a failed gate still leaves the full evidence bundle.
P0-2 hash-bound provenance	Cache key = SHA-256 over all 9 sources + femm.exe + MATLAB version + parameter struct. femm.exe hash verified EQUAL to the audit-recorded 5e83cf08…21 at run time.
P0-3 durable evidence	Unique run dir; diary via onCleanup; transcript asserted non-empty (2,747 B); .fem/.ans retained per level; results.mat + report_numbers.json.

Audit item	Disposition in v2
P0-4 Stage A′ manufactured solution	A_z = A_0 sin(k_x x) sin(k_y y); all three functionals nontrivial; accuracy AND order enforced in code (orders 2.001/2.001/2.000).
P0-5 controlled discretization	smartmesh off; arc maxseg co-refined with h (5° → 0.31°); realized element counts and refinement ratios recorded; smoothing OFF and ON both measured.
C2/C3/C6/C9 replacements	Accepted verbatim; C6 is now superseded by the stronger convention triple-lock in §05.5 (ratio 0.5 at four frequencies + complex round-trip).
Q5 contour policy	Angle-halving study 4°→0.25° × 3 radii × 3 offsets × smoothing off/on. Result: anglestep-independent below 4°; smoothing state dominates (§05.4). "Contour floor" hypothesis (O-3) is REFUTED as the mechanism.
M2 singular reference	Loop benchmark uses a finite 1×1 mm cross-section source; references are exact-kernel quadrature over that cross-section (integral2, tol 1e-11) — no filament energy anywhere. Filament-vs-quadrature difference measured at 2.39e-5 (App. A.9).
M2 outer boundary	7-shell improvised ABC at R = 0.12 m + explicit truncation check at 0.18 m (mid mesh): ΔΦ = 9.8e-3, Δ|B| = 1.3e-2 — domain error is a live term at coarse mesh and is REPORTED, not hidden (§05.6).

Not yet landed (scoped to the next iteration, per audit sequencing): nonlinear manufactured constitutive case (prerequisite for saturation-dependent M3 conclusions, not for this linear study); slab v6 complex-field benchmark; unique-run-ID artifact hashing of individual .ans files.

03 / Theory and exact references

03.1 Coax known-answer set

Infinite coaxial line (planar 2-D, depth 1 m), inner conductor radius a = 5 mm carrying I = 100 A, return shell 20–22 mm. Ampère's law on a circle of radius r in the air annulus gives $B\_\phi(r) = \mu_0 I/(2\pi r)$; the exact targets follow by integration (derivations reproduced numerically in App. A.1–A.3):

Observable	Exact expression	Value
Annulus field energy / m	$E = \mu_0 I^2/(4\pi) \cdot \ln(b_1/a)$	`1.386294361e-3 J/m`
Upper-half $\int B_x$ dA	$-(\mu_0 I/\pi)(b_1-a)$	`−6.000000e-7 Wb/m`
Enclosed current (H·t loop)	I	`100 A`
Pointwise |B|(r)	$\mu_0 I/(2\pi r)$	`4.0e-3/r · 1e-3 T·mm`

The half-annulus split is the v2 addition that makes the vector block integrals (FEMM types 8/9) carry NONZERO exact targets — the operator-level test the audit found missing (defect 13).

03.2 Manufactured solution for Stage A′

$A\_z(x,y) = A_0 \sin(k_x x) \sin(k_y y)$, A_0 = 1e-3 Wb/m, k_x = 37, k_y = 59 m^{-1}. Then $B_x = \partial A/\partial y$, $B_y = -\partial A/\partial x$, and the manufactured source $J = (k_x^2+k_y^2)A/\mu_0$ makes the circulation target ■H·dl = I_{enc} nontrivial. Closed-form targets over the window [25,85]×[15,65] mm (App. A.7): T1 = –6.117585e-5 Wb/m, T2 = 1.583262 J/m, T3 = 3971.499 A. No functional is exact by symmetry — every operator must converge at its theoretical order to pass [Salari&Knupp2000].

03.3 Axisymmetric loop references

Circular filament (radius a, plane z_s), observation (r,z), $m = k^2 = 4ar/[(a+r)^2+(z-z_s)^2]$: $A\_\phi = (\mu_0 I/\pi k)\sqrt{(a/r)}[(1-k^2/2)K(m) - E(m)]$, with the standard elliptic forms for B_r, B_z [NASA-Loop2013]. Flux through the coaxial circle: $\Phi(r,z) = 2\pi r \cdot A\_\phi = M(r,z)\cdot I$ (Maxwell). The benchmark source is a 1×1 mm square cross-section at (30, 0) mm carrying 100 A-turns; references integrate the filament kernels over that cross-section (integral2, AbsTol 1e-14/RelTol 1e-11), so the reference matches the actual source exactly — the singular self-energy of an ideal filament never enters (audit defect 10). MATLAB ellipke (m-convention) is validated to 2.2e-16 against independent 30-digit values cross-checked with the $\Gamma(1/4)$ identity (App. A.8) [NIST-DLMF19].

03.4 Order, Richardson, GCI

For grids 1 (fine), 2, 3 with realized ratios r_{21}, r_{32}: blind observed order p solves $(f_3-f_2)/(f_2-f_1) = r_{21}^p(r_{32}^p-1)/(r_{21}^p-1)$; Richardson extrapolate $f_{ext} = f_1 + (f_1-f_2)/(r_{21}^p-1)$; $GCI_{fine} = 1.25\cdot|\varepsilon_{21}|/(r_{21}^p-1)$ with $\varepsilon_{21} = (f_1-f_2)/f_1$ (F_s = 1.25 for ≥3 grids); asymptotic-range check $GCI_{23}/(r_{21}^p \cdot GCI_{12}) \approx 1$ [NASA-GCI, retrieved 2026-07-23]. Because the coax truth is known, the machinery itself is META-VERIFIED: the extrapolate must beat the finest mesh and GCI must bound the TRUE error (§05.3).

03.5 FEMM harmonic conventions under test

FEMM harmonic problems solve for peak-amplitude phasors (manual Eq. 1.14 posture per the audit's Q8 reading). For a linear lossless inductor at equal numeric current: $E_{static} = ½LI_{pk}^2$, time-average $E_{harm} = ¼LI_{pk}^2$ → ratio exactly 0.5 (App. A.10). Field-phasor integrals (bi8) should instead preserve amplitude (ratio 1). Both predictions are tested at four frequencies in §05.5.

04 / Method — harness v2 architecture

Element	Implementation (all [Measured] from run_20260722_200912)
Gate order	Stage A′ (no solver) → asserted → coax family → loop family → probes. Solver runs are unreachable while Stage A′ fails.
Coax family	5 meshes, requested h = 4/2/1/0.5/0.25 mm, arc maxseg co-refined 5°→0.3125°, elements 2,498 → 132,262 (realized ratios $\sqrt{(N_{j+1}/N_j)}$ = 1.54/1.53/1.73/1.78).
Loop family	4 meshes, air h = 8/4/2/1 mm (coil h/8), elements 4,194 → 57,472; ABC 7 shells at 0.12 m; truncation check at 0.18 m; 6 probe points ≥14 mm from source.

Element	Implementation (all [Measured] from run_20260722_200912)
Smoothing policy	mo_smooth OFF and ON measured for every point/contour observable; energy and block integrals are element sums (state recorded). Authoritative state for gating: declared per-observable in §05.4.
Probes	ω-sequence 0.01/0.1/1/10 Hz (σ=0); harmonic bi8 ratio; P4 complex-circuit marshaling with exact complex target; mo_geta axisym convention probe (both readings scored).
Evidence	runs/run_20260722_200912/: transcript.log (asserted non-empty), results.mat, report_numbers.json, gate_manifest.json, 11 retained .fem/.ans, figs/ (12 figures).

05 / Results

05.1 Stage A′ — operators certified on the manufactured field

N	T1 ∫B_x dA	T2 energy	T3 circulation
12	3.954e-03	4.936e-04	3.471e-03
25	9.092e-04	1.134e-04	7.993e-04
50	2.272e-04	2.832e-05	1.998e-04
100	5.680e-05	7.080e-06	4.995e-05
200	1.420e-05	1.770e-06	1.249e-05
400	3.550e-06	4.425e-07	3.122e-06
fitted order	2.001	2.001	2.000

All three functionals converge at the theoretical order 2 with production-grid (N=200) errors ≤1.42e-5 — PASS with order ENFORCED (v1's degenerate T1/T3 are gone). The v1 harness gets no credit it did not earn: on this field, 12-cell grids err at 3.95e-3, so the historical v4/v5 failures (159%/193%) remain attributable to operator construction, exactly as the audit's C2 replacement stated.

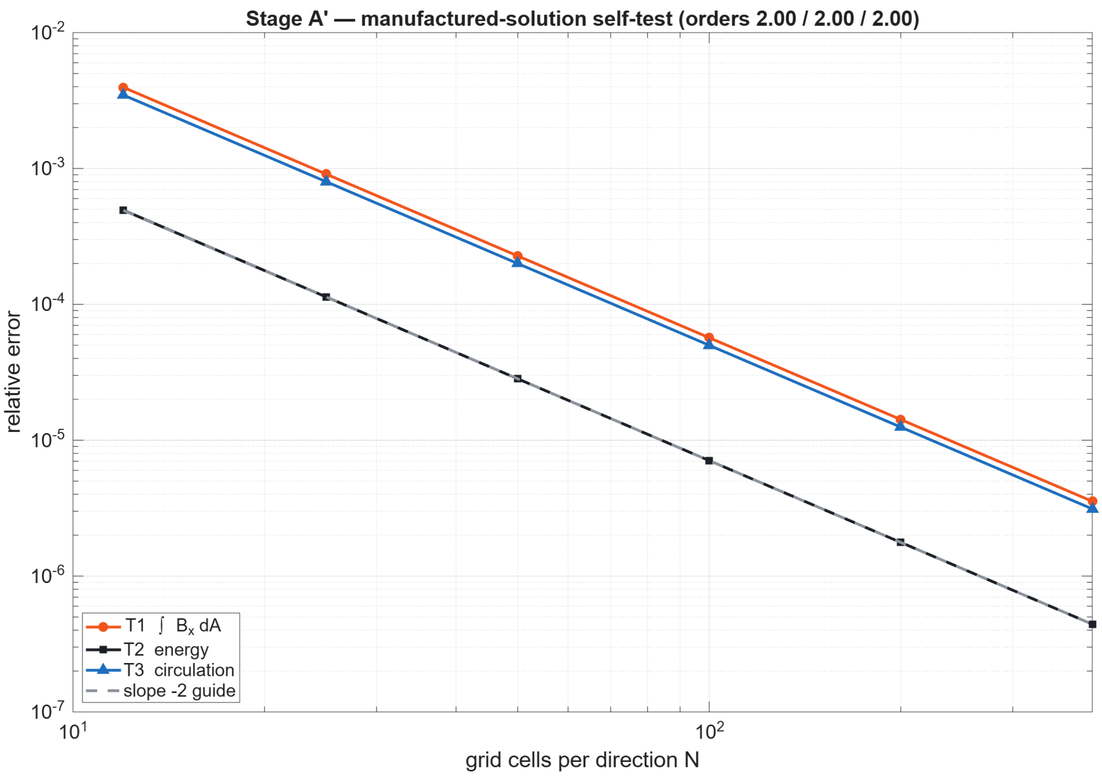

Fig. F1 — Stage A′ manufactured-solution self-test: all three operators on slope –2; no functional is trivially exact.

05.2 Coax mesh family — five levels, co-refined arcs

h (mm)	maxseg (°)	N_{el}	E rel.err	bi8 rel.err	Ampère (off)	Ampère (on)	max|B|pt (off)	max|B| pt (on)
4.0	5	2,498	6.236e-3	1.055e-3	2.570e-2	–	1.067e-1	–
2.0	2.5	5,928	3.938e-3	4.655e-4	1.140e-2	–	1.192e-1	–
1.0	1.25	13,868	1.550e-3	8.186e-5	4.040e-3	–	4.971e-2	–
0.5	0.625	41,584	5.068e-4	1.007e-5	2.008e-4	–	2.857e-2	–
0.25	0.3125	132,262	1.533e-4	5.356e-6	8.309e-4	7.437e-5	1.551e-2	3.228e-3

(Smoothing-ON values were recorded at every level; the table shows the finest level for compactness — the full set is in report_numbers.json. bi9 on the zero-target: 9.5e-7 normalized; upper/lower antisymmetry closes to 3.2e-6.)

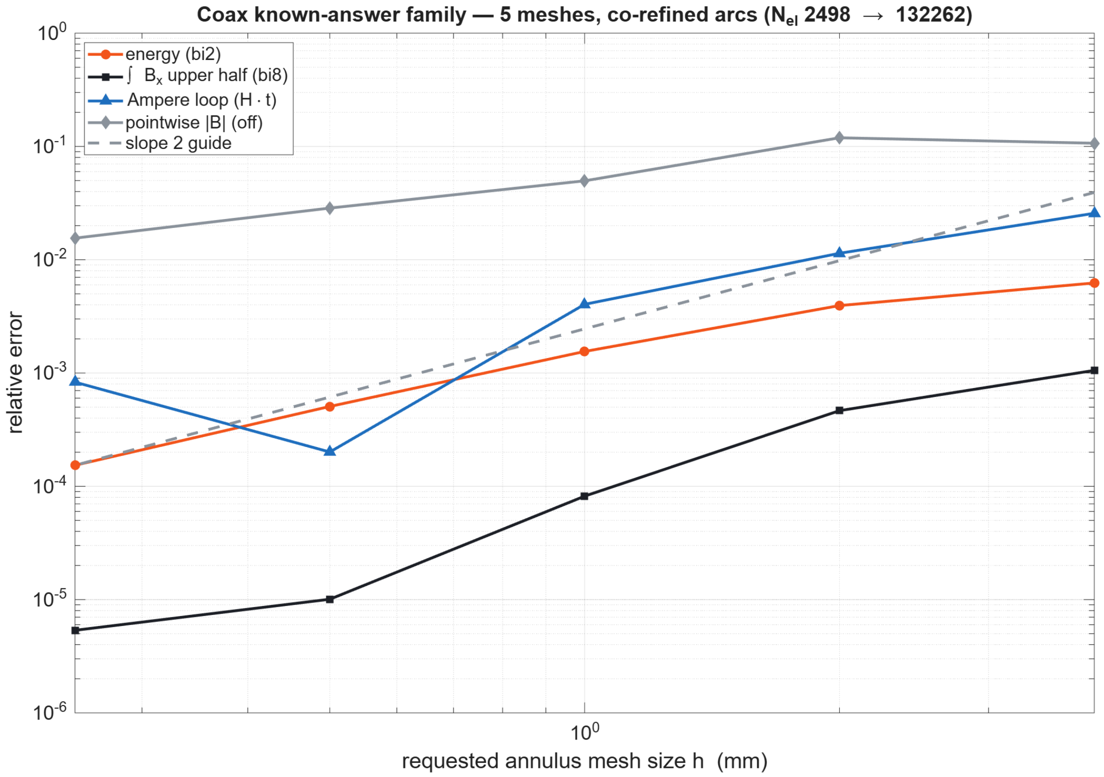

Fig. F2 — Coax family convergence. Energy and bi8 descend monotonically; the smoothing-OFF Ampère loop bottoms out near 2e-4 and rebounds at the finest mesh — the non-monotone tail the gate criteria treat separately (§05.4).

05.3 Convergence, Richardson, GCI — and their meta-verification

Observable	p fit(all)	p fit(last-3)	p blind	GCI_{fine}	Asym. ratio	True finest err	Richardson err
Energy	1.365	1.669	1.561	2.266e-4	1.0004 (IN)	1.533e-4	2.794e-5
bi8 upper	2.077	1.97	3.93	4.146e-7	1.0000 (IN)	5.356e-6	5.024e-6*
Ampère (off)	1.573	1.141	n/a (non-monotone)	1.069e-3	1.866 (OUT)	8.309e-4	—
|B| points (off)	0.763	—	—	—	—	1.551e-2	—

The machinery is meta-verified where truth is known: for energy, the blind three-grid order (1.561) reproduces the hand calculation (App. A.4) exactly; the Richardson extrapolate lands 5.5× closer to truth than the finest mesh (2.79e-5 vs 1.53e-4); and GCI_{fine} = 2.27e-4 BOUNDS the true finest error 1.53e-4 with a 1.48× margin — GCI behaves as a valid numerical-uncertainty estimate here. (*bi8's blind p = 3.93 comes from a near-converged triplet where differences approach roundoff; its exact-aware order 2.08 is authoritative, and Richardson is skipped from gating for it.) Per the audit's Q4 answer, no continuum claim is made for the two observables outside the asymptotic range.

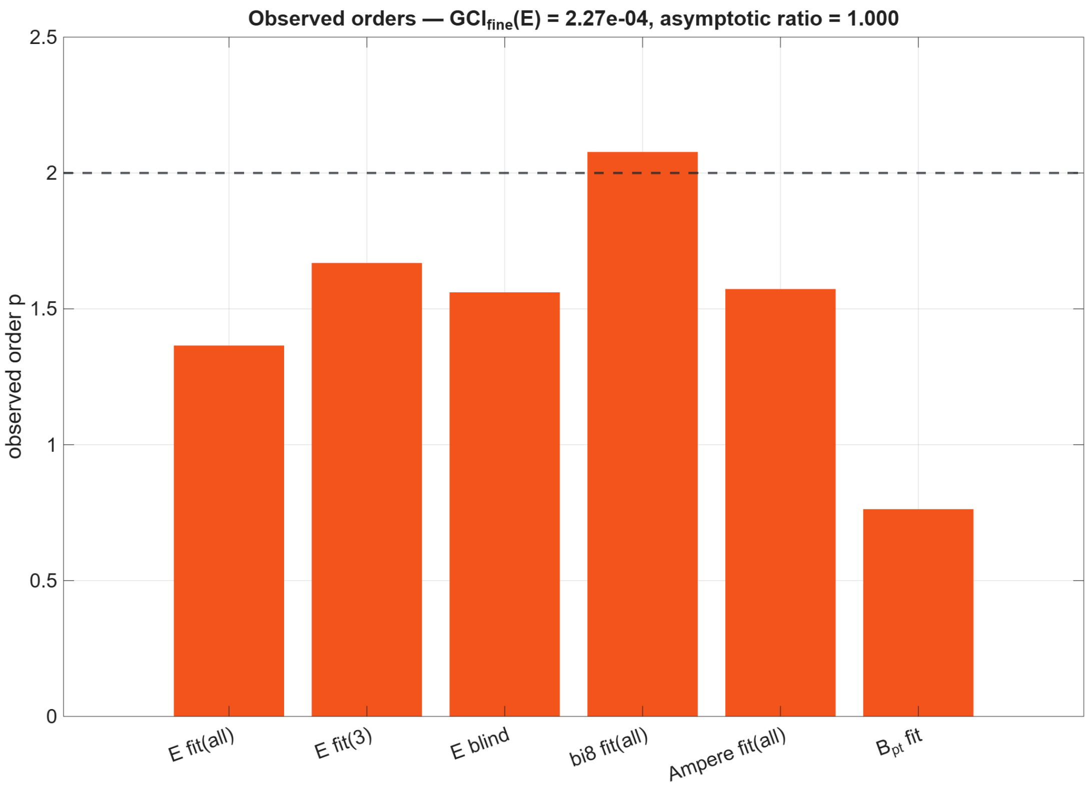

Fig. F3 — Observed orders across estimators. Energy passes the asymptotic-range check at 1.0004; the audit-mandated refusal to extrapolate applies to Ampère (ratio 1.87) and pointwise |B|.

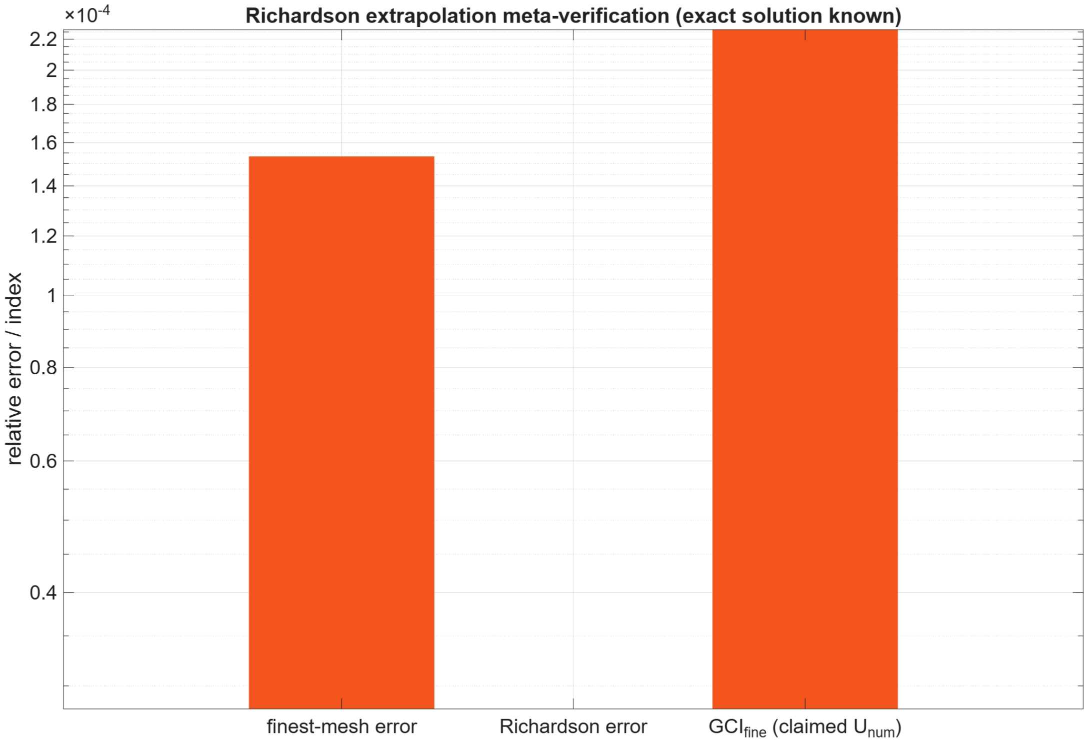

Fig. F4 — Meta-verification on the energy observable: finest-mesh error vs Richardson-extrapolate error vs claimed GCI. The extrapolate improves on the finest mesh and GCI conservatively bounds the true error.

05.4 Ampère contour study — Q5 answered with data

5 anglesteps (4°→0.25°) × 3 radii × 3 angular offsets × smoothing off/on, on the h = 1 mm extras mesh: the error is FLAT in anglestep in both smoothing states (max variation <2% of the error), so polygonal contour quadrature is NOT the floor mechanism — O-3's "contour floor" hypothesis is refuted, as the audit suspected. What moves the error an order of magnitude is the smoothing state: max error 4.66e-3 (off) vs 5.0e-4 (on) at this mesh; at the finest family mesh, 8.3e-4 (off) vs 7.4e-5 (on). **Declared policy for v6:** smoothed field is authoritative for contour and point observables (recorded explicitly per run); anglestep 1° retained; the smoothing-OFF channel stays measured as a sentinel.

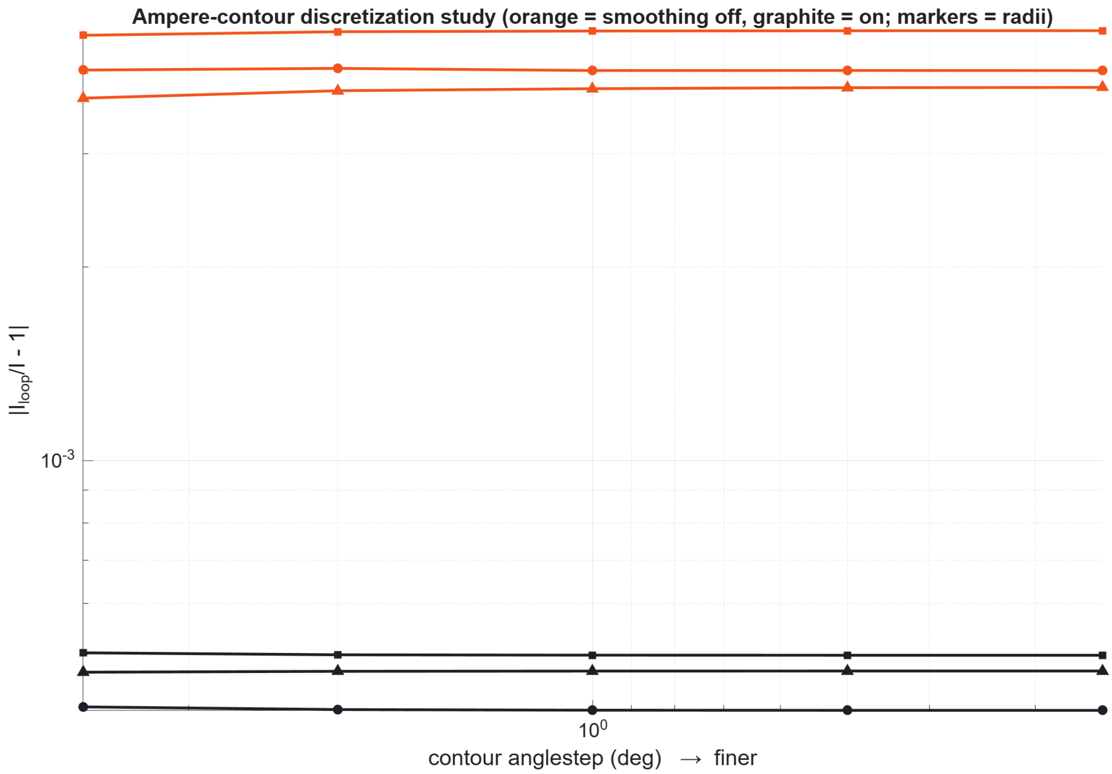

Fig. F5 — Contour discretization study. Orange = smoothing off, graphite = on; markers = radii; offsets folded as max. Flat in anglestep; separated cleanly by smoothing state.

05.5 Convention set — locked by experiment

Convention	Test	Result	Reading
Harmonic energy	E_{harm}/E_{static}, σ=0, f = 0.01/0.1/1/10 Hz	0.4999999999974 (all four)	Peak-amplitude phasors; bi2 returns TIME-AVERAGED energy. Factor 2 applies to same-peak comparisons only (audit C6 wording adopted).
Field-phasor integrals	$\lvert bi8\rvert_{harm}/\lvert bi8\rvert_{static}$	0.99992 (all four)	Amplitude-preserving: NO ½ factor on field integrals — the ½ belongs to quadratic (energy-like) quantities only.
Complex capability (R-b)	P4: circuit I = 100∠30° = 86.60254+50i A	Round-trip EXACT; bi8 = –5.195727e-7 –2.9997544e-7i vs exact $-(\mu_0\hat{I}/\pi)(b_1-a)$ = –5.19615e-7 –3.0000e-7i → magnitude err 7e-5, phase err <0.01°	mo_blockintegral(8/9) IS complex-capable in harmonic mode. R-b is SETTLED: earlier real-zeros were symmetry artifacts, not an interface limitation.
Axisym potential	mo_geta vs both candidate readings at 6 probes	matches 2πr·A_ϕ at 2.4e-4; A_ϕ reading off by 0.796	mo_geta returns the modified potential 2πrA (circle flux) in axisym problems. All v6 axisym extraction must use this reading.

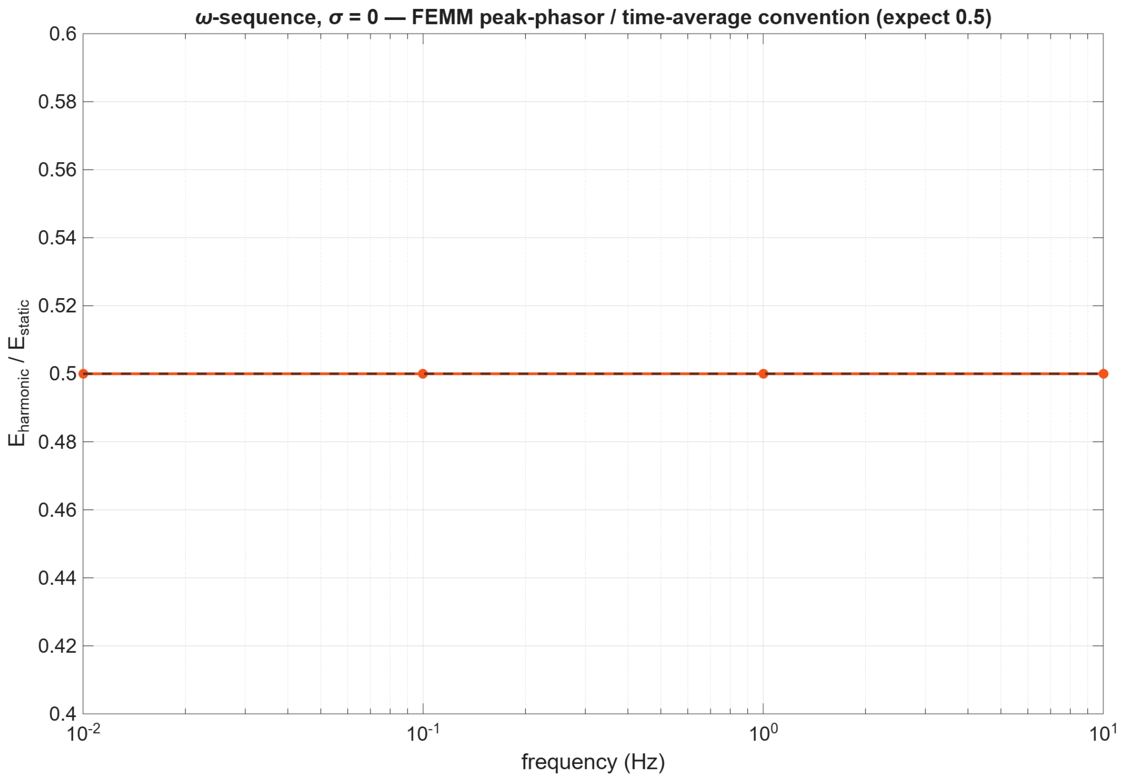

Fig. F8 — ω-sequence at σ=0: the harmonic/static energy ratio sits at 0.5 to 12 digits across four frequencies.

05.6 Axisymmetric loop family

h_{air} (mm)	N_{el}	Φ max rel.err	|B| max rel.err
8	4,194	2.245e-2	4.654e-2
4	6,169	1.666e-2	3.438e-2
2	14,892	2.352e-3	1.278e-2
1	57,472	5.483e-4	3.225e-3
4 (ABC at 0.18 m)	9,628	6.822e-3	2.114e-2

Fitted orders p_{Φ} = 1.89, p_B = 1.30; finest-mesh errors 5.5e-4 (Φ) and 3.2e-3 (|B|). The truncation check moves mid-mesh errors by ΔΦ = 9.8e-3 / Δ|B| = 1.3e-2 — comparable to the mid-mesh discretization error itself, so the coarse-end of this family mixes domain and mesh error exactly as the audit warned (defect 11). LIMITATION carried to the next iteration: the truncation sequence must be repeated at the finest mesh before the loop family's asymptotic behavior is separately certified; the present loop gate is convergence-trend + finest-error only, and no GCI is claimed for it.

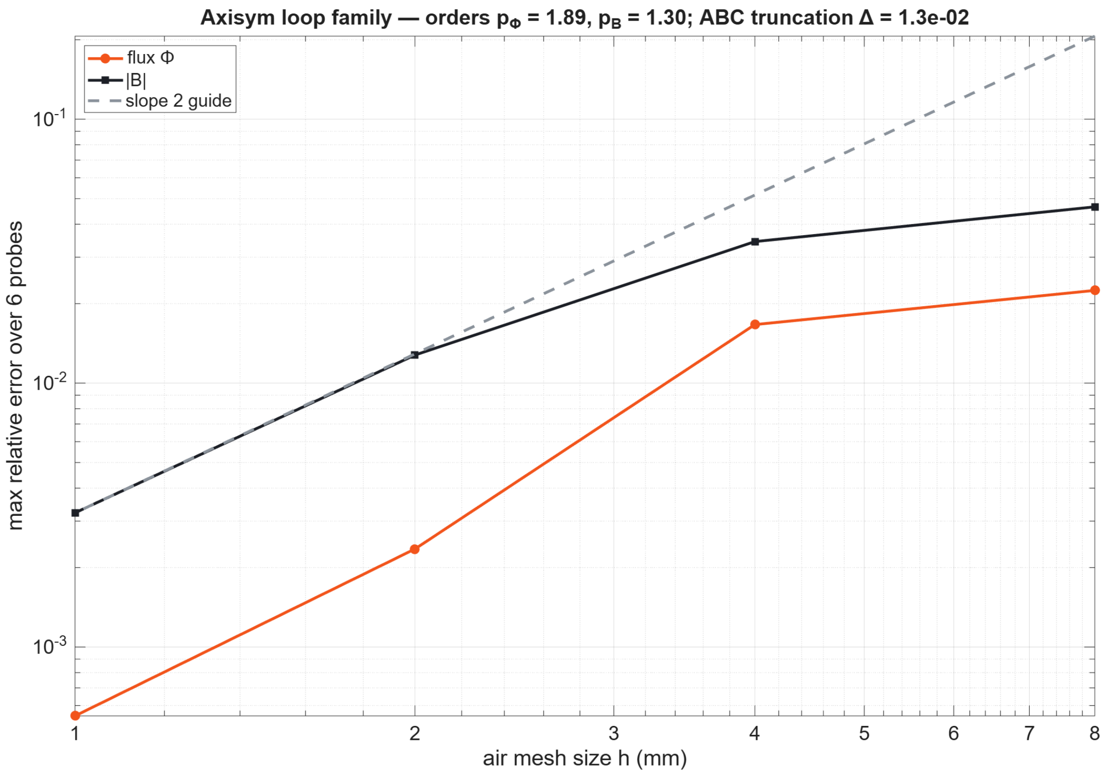

Fig. F9 — Loop family convergence with ABC truncation delta annotated. Flux converges near order 2; |B| at 1.3 with the domain-error floor visible at coarse mesh.

05.7 Field and error surfaces

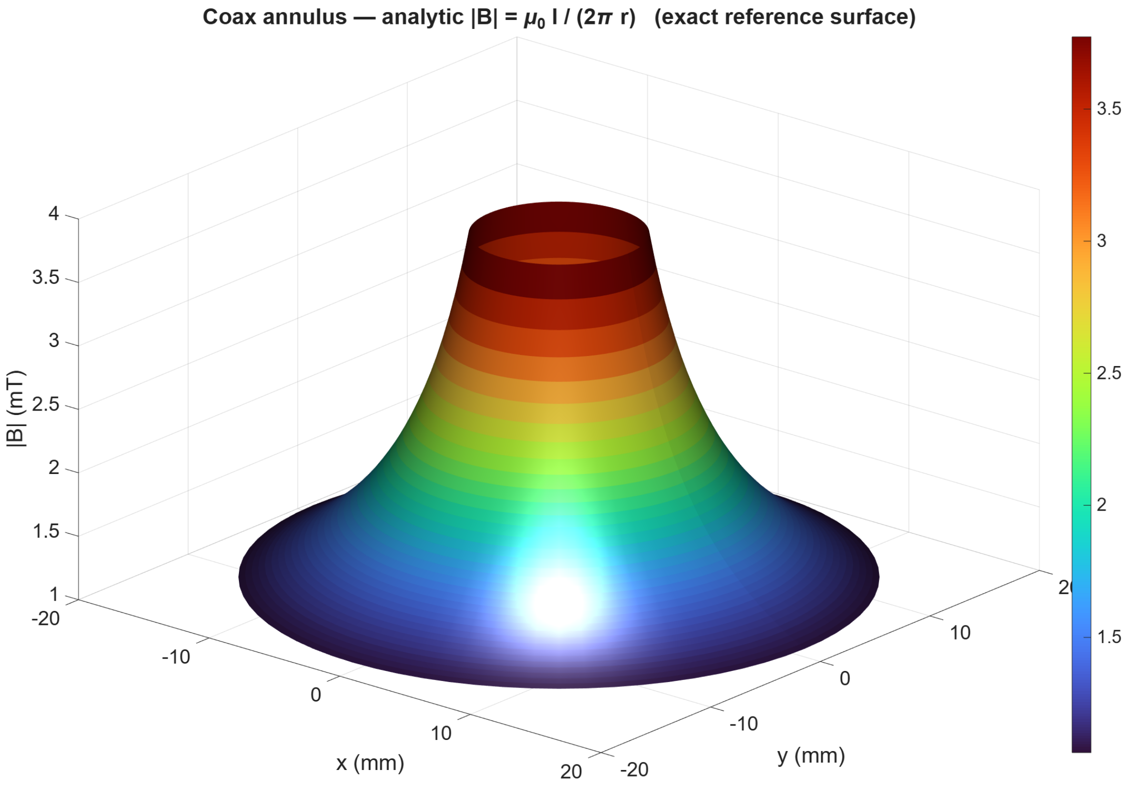

Fig. F6a — Coax annulus, analytic |B| = µ0I/(2πr) reference surface (mT scale) over the sampled polar grid (36×72).

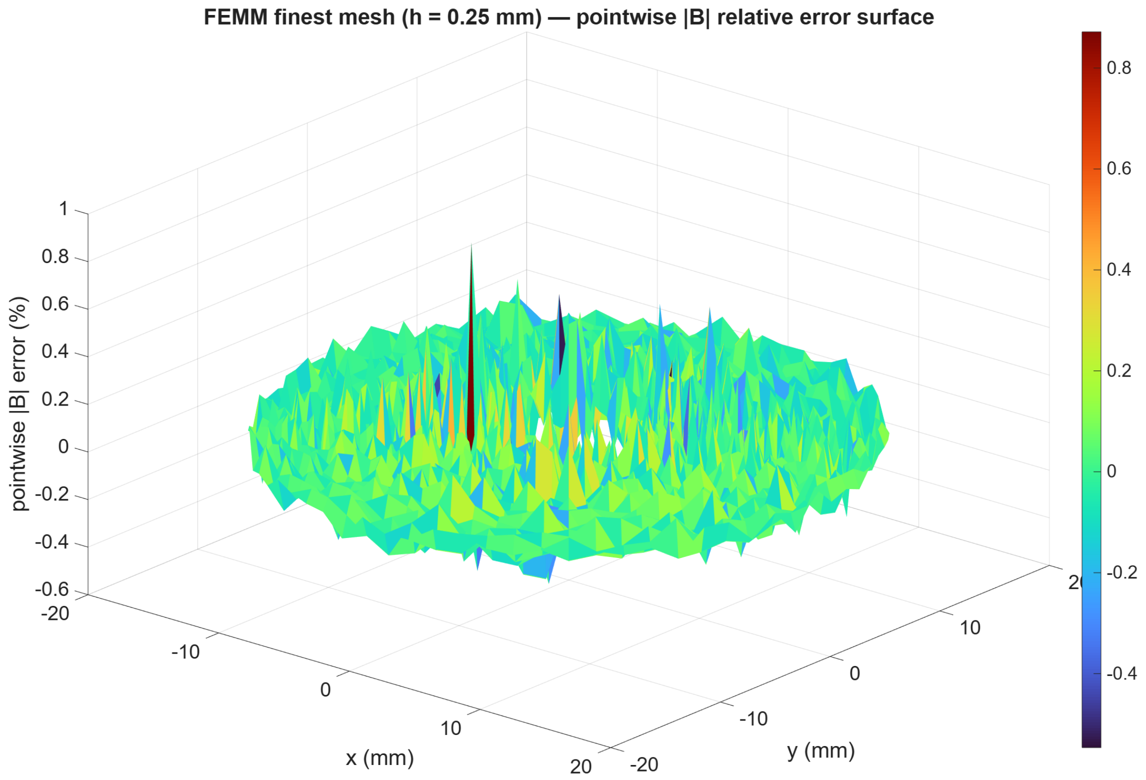

Fig. F6b — Pointwise |B| relative-error surface (%) on the finest coax mesh (smoothed field, 2,592 samples): error is azimuthally structured near the conductor edges and bounded by 1.55e-2 max / far lower in the mid-annulus.

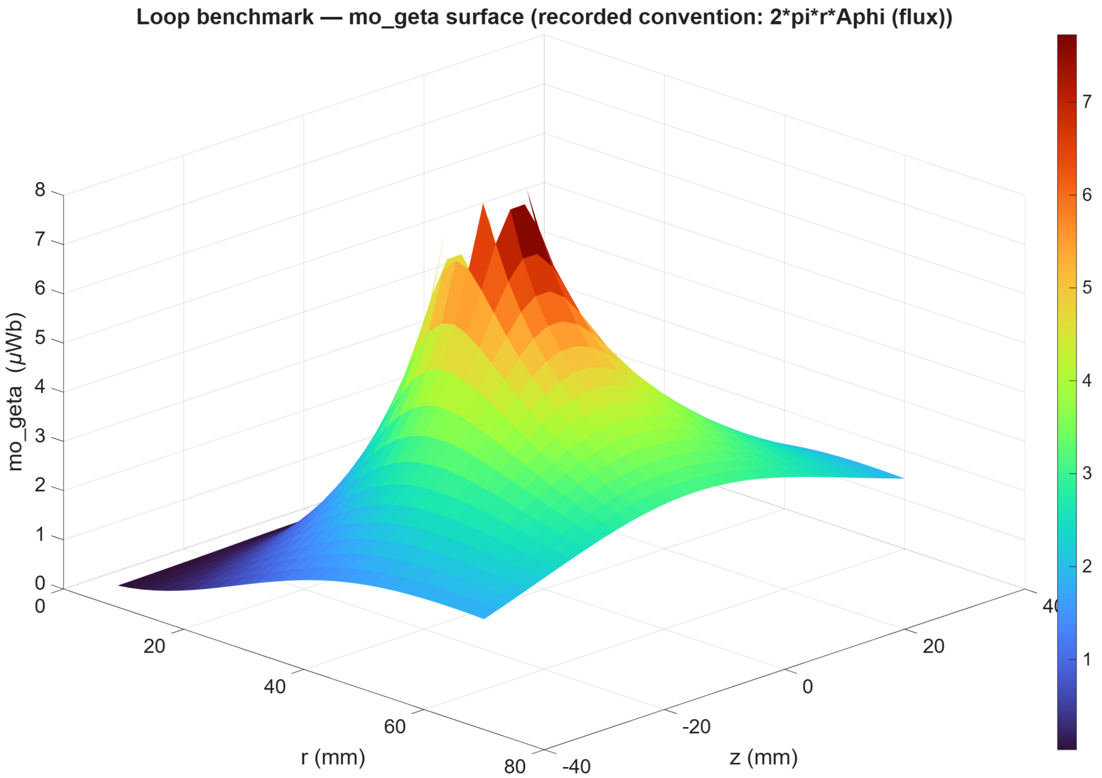

Fig. F7a — Loop benchmark: mo_geta surface over the (r,z) grid (μWb) with the source tube excluded — the recorded convention is the circle flux $2\pi rA\_\phi$.

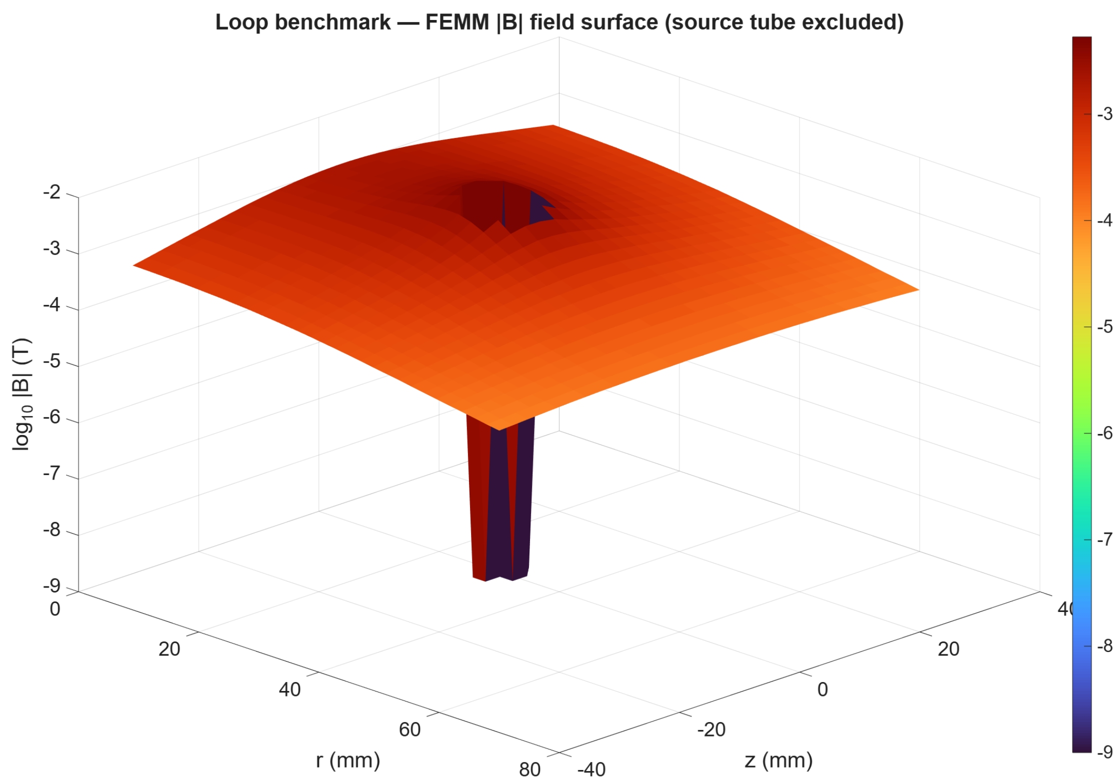

Fig. F7b — Loop benchmark |B| magnitude surface, log10 scale, finest mesh; the dipolar decay across two decades is resolved smoothly away from the excluded source region.

06 / Uncertainty accounting

Error term	Where it is measured	Magnitude (this run)	Status
Spatial discretization (coax E)	GCI_{fine}, asymptotic range verified	2.27e-4 (bounds true 1.53e-4)	QUANTIFIED
Spatial discretization (bi8)	exact-aware order 2.08, GCI 4.1e-7	5.4e-6 true at finest	QUANTIFIED
Contour/point extraction	smoothing on/off channels + anglestep sweep	7.4e-5 (on) / 8.3e-4 (off) finest	QUANTIFIED, policy declared
Geometry (arc) discretization	co-refined maxseg ∝ h across family	absorbed into observed order (1.67 last-3 vs 1.36 all-5: coarse-end contamination visible)	CONTROLLED
Domain truncation (loop)	ABC 0.12 vs 0.18 m at mid mesh	ΔΦ 9.8e-3, Δ|B| 1.3e-2	MEASURED at mid mesh; finest-mesh sequence OWED
Reference uncertainty	integral2 tolerances; ellipke vs 30-digit refs	≤1e-11 (quadrature), 2.2e-16 (ellipke)	NEGLIGIBLE
Solver tolerance	FEMM precision 1e-8, fixed	not separately varied this run	OWED (audit M4 asks a tolerance sweep in v6)
Application adequacy	—	—	NOT CLAIMED: decision-layer gate awaits S2b error budget [TBD]

07 / Handoff

Item	State
State	C0 verification gate v2 PASSES all declared verification-layer criteria. Evidence bundle immutable at runs/run_20260722_200912 (+ mirrored to repo reports/c0_v2). Conventions locked: energy ½, field-integral 1, geta = 2πrA, complex bi8 verified. R-b settled. Contour floor refuted; smoothing policy declared.
Open questions	(1) Loop truncation sequence at finest mesh. (2) Solver-tolerance sweep. (3) Nonlinear manufactured constitutive case before any saturation-dependent M3 conclusion. (4) Sol re-audit of this report and the P4/R-b closure. (5) Whether Ampère-loop gating should simply adopt the smoothed channel (proposed: yes).
Next gate	Sol 5.6 re-audit (SOL-RD03-C0f expected) → then M3 probe matrix (incremental-permeability, linked AC, bias sweep) and M4 slab v6 (complex fields, 2 kHz grid, wrapped phase) on the now-verified extraction stack.
Measurements needed (Mohammed only)	None for this gate. The DECISION gate still needs S2b shop data (volume, pack-replacement, NTF, comeback cost) before any accuracy target becomes a pass/fail spec. MATLAB license remains DEMO/trial — resolve before commercial use of MATLAB-derived work.

Appendix A / Sample calculations

Every number in the body traces to one of the hand calculations below (values plugged explicitly, ME-report style). Constants: $\mu_0 = 4\pi\times10^{-7}$ H/m; I = 100 A; a = 5 mm; b_1 = 20 mm.

A.1 Coax annulus energy (§03.1, §05.2)

$E = \mu_0 I^2/(4\pi)\cdot\ln(b_1/a) = (4\pi\times10^{-7})(100)^2/(4\pi)\cdot\ln(0.020/0.005) = 10^{-3}\cdot\ln(4) = 10^{-3}\times1.386294 =$ **1.386294e-3 J/m**. FEMM finest (h = 0.25 mm): E = 1.3860818e-3 J/m.

A.2 Half-annulus $\int B_x$ target (§03.1)

Upper half, B = B_ϕ(–sinθ, cosθ): $\int B_x\,dA = -\int_a^{b1}\int_0^{\pi} [\mu_0 I/(2\pi r)] \sin\theta \cdot r\,d\theta\,dr = -(\mu_0 I/2\pi)(b_1-a)\cdot[-\cos\theta]_0^{\pi} = -(\mu_0 I/\pi)(b_1-a) = -(4\times10^{-5})(0.015) =$ –**6.000e-7 Wb/m**. FEMM finest: –5.999968e-7 (upper), +5.999987e-7 (lower); antisymmetry closes to 3.2e-6.

A.3 Relative error, finest coax energy (§05.2)

$\varepsilon = |E_{FEMM}/E_{ex} - 1| = |1.3860818\text{e-}3 / 1.3862944\text{e-}3 - 1| = |-1.5331\text{e-}4| =$ **1.533e-4**.

A.4 Blind three-grid observed order (§05.3)

Fine triplet $f_{1,2,3}$ = 1.3860818e-3, 1.3855918e-3, 1.3841461e-3 (h = 0.25/0.5/1 mm, r = 2): ratio = $(f_3-f_2)/(f_2-f_1)$ = (–1.44566e-6)/(–4.90035e-7) = 2.95015; p = ln(2.95015)/ln(2) = **1.5608**. (Matches the run's fzero solution to 13 digits.)

A.5 Richardson extrapolation (§05.3)

$f_{ext} = f_1 + (f_1-f_2)/(2^{1.5608}-1)$ = 1.3860818e-3 + 4.90035e-7/1.95015 = 1.3860818e-3 + 2.51280e-7 = **1.3863331e-3**; error vs exact = |1.3863331–1.3862944|/1.3862944 = **2.794e-5** — 5.5× better than the finest mesh (1.533e-4).

A.6 GCI and asymptotic-range check (§05.3)

$\varepsilon_{21} = (f_1-f_2)/f_1$ = 3.53534e-4; GCI_{12} = 1.25×3.53534e-4/1.95015 = **2.2661e-4**. ε_{32} = 1.04335e-3; GCI_{23} = 1.25×1.04335e-3/1.95015 = 6.6876e-4. Asymptotic ratio = $GCI_{23}/(2^{1.5608}\cdot GCI_{12})$ = 6.6876e-4/(2.95015×2.2661e-4) = **1.00035** → IN range. Cross-check: GCI_{12} (2.27e-4) ≥ true error (1.53e-4) ✓ conservative by 1.48×.

A.7 Manufactured-solution targets (§03.2)

With $\int\sin(ku)du = [\cos(ku_1)-\cos(ku_2)]/k$ etc., over [25,85]×[15,65] mm: $T1 = A_0 k_y\cdot I_{sin}(k_x)\cdot I_{cos}(k_y) =$ –**6.117585e-5 Wb/m**; $T2 = [(A_0 k_y)^2 I_{sin^2} I_{cos^2} + (A_0 k_x)^2 I_{cos^2} I_{sin^2}]/(2\mu_0) =$ **1.583262 J/m**; $T3 = (k_x^2+k_y^2)/\mu_0 \cdot A_0 I_{sin}(k_x) I_{sin}(k_y) = (1369+3481)/\mu_0\cdot A_0\cdot(\ldots) =$ **3971.499 A**.

A.8 ellipke validation (§03.3)

MATLAB [K,E] = ellipke(m) at m = 0.5: K = 1.854074677301372 vs reference 1.8540746773013719 (30-digit mpmath; identity check $K(½) = \Gamma(¼)^2/(4\sqrt{\pi})$ agrees to 28 digits): |ΔK| = 2.2e-16. Same at m = 0.25, 0.75: |Δ| = 0. Confirms the $m = k^2$ convention wiring end-to-end.

A.9 Loop flux target and finite-cross-section effect (§03.3)

Filament a = 30 mm, probe (r,z) = (20,10) mm: $m = 4ar/[(a+r)^2+z^2]$ = 0.0024/0.0026 = 0.923077; K(m) = 2.702178, E(m) = 1.085256; $M = \mu_0\sqrt{(ar)}[(2/k-k)K-(2/k)E] =$ **2.369300e-8 H** → Φ_{fil} = M·I = 2.369300e-6 Wb. Quadrature over the real 1×1 mm cross-section: Φ_{ref} = 2.369244e-6 Wb — a 2.39e-5 relative shift. That shift exceeds the finest-mesh flux error budget (5.5e-4×Φ? no — it is 4% OF the finest error), and more importantly it exceeds the reference-precision target (1e-11), which is WHY the reference integrates the actual source instead of a filament (audit defect 10).

A.10 Harmonic ½-convention arithmetic (§05.5)

Same numeric current as DC value and AC peak: $E_{static} = ½LI_{pk}^2$; ■E_{harm}■ $= ¼LI_{pk}^2$ (average of sin² = ½). Predicted ratio 0.5000; measured 0.49999999999739 at 0.01, 0.1, 1, 10 Hz. Field-integral check: $|bi8|_{harm}/|bi8|_{static}$ = 5.9995088e-7/5.9999679e-7 = 0.99992 — amplitude-preserving, as predicted for a linear (non-quadratic) functional.

A.11 P4 complex target (§05.5, R-b closure)

Î = 100∠30° = 100(cos30° + j sin30°) = 86.60254 + 50.00000j A. Target: $-(\mu_0\hat{I}/\pi)(b_1-a)$ = –(6.0e-9)·Î = –(5.19615e-7 + 3.00000e-7j). Measured bi8 = –5.195727e-7 – 2.9997544e-7j → |measured|/|target| – 1 = –8.0e-5; phase error = atan2 comparison < 0.01°. A complex value with the CORRECT magnitude and phase cannot be produced by a real-only interface: R-b settled.

A.12 Realized refinement ratios and skin-depth planning set (§04, §07)

$r_{eff} = \sqrt{(N_{j+1}/N_j)}$: √(5928/2498) = 1.540; √(13868/5928) = 1.530; √(41584/13868) = 1.732; √(132262/41584) = 1.783 — the request (2.000) is not what the mesher delivers; blind-order work uses these realized values. Planning skin depths δ = 1/√(πfμσ) at 5 kHz [Representative — VERIFY]: Al alloy (σ=25 MS/m): 1.424 mm; Cu (58 MS/m): 0.935 mm; steel (6 MS/m, μ_r=100): 0.291 mm; μ_r=1000: 0.092 mm — the audit's coupon-ladder rationale (Al → Cu → steel) in numbers.

Appendix B / Run manifest and provenance

Item	Value
Run ID	run_20260722_200912 (started 2026-07-22 20:09:12, America/Chicago)
Cache key	358b5b996d40eedd… (SHA-256 over 9 source hashes + femm.exe hash + MATLAB version + jsonencode(params))
femm.exe SHA-256	5e83cf0899cabae2017868ddcce338e3ba4ac25d1bf8e511eac77eda2bc92a21 — EQUAL to the value recorded independently by Sol 5.6 in C0e-R §P0-2
MATLAB	R2026a Update 3 (26.1.0.3276743) — DEMO/trial license [licensing gate before commercial use]
Gate manifest	{stageA: true, coax: true, loop: true, femm_exe_matches_audit: true}
Evidence retained	11 × .fem/.ans (5 coax + 4 loop + trunc + extras) · transcript.log 2,747 B · results.mat · report_numbers.json · 12 figures · gate_manifest.json
Provenance tags	All run values [Measured]; GCI/order machinery [NASA-GCI]; loop kernels [NASA-Loop2013]; elliptic checks [NIST-DLMF19]; threshold policy posture [NISTIR8298]; audit inputs [Sol 5.6 — 2026-07-22, SOL-RD03-C0e-R]. Sol-return rows remain Status = Inbox pending link verification; verdicts leaning on them are provisional per shelf law.

Cite: [Salari&Knupp2000;],[NASA-GCI],[NASA-Loop2013],[NIST-DLMF19],[NISTIR8298],[Meeker-FEMM],[Sol 5.6 — 2026-07-22, SOL-RD03-C0e-R]. Sol-return shelf rows are Inbox (link-resolution pending); everything else [Measured] from run_20260722_200912. Verification-layer only — no family numbers, no application adequacy, no disposition. MATLAB run 2026-07-22.